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Modeling the evolution of collective overreaction in dynamic online product diffusion networks

Xiaochao Wei [a,*], Yanfei Zhang [a], Xin (Robert) Luo [b]

[a] Department of Economy, Wuhan University of Technology, China
[b] Anderson School of Management, The University of New Mexico, USA

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ABSTRACT

With the development of e-commerce, collective overreactions such as buying frenzy have become prominent. However, studies have rarely investigated the mechanism of irrational consumer behavior at the group level. To investigate the evolution of collective overreaction in dynamic online product diffusion networks, we employed a sequential multiple-methods approach. A conceptual model is constructed to capture the influence of social network dynamic evolution on individual irrationality. An agent-based model (ABM) under different network dynamic growth mechanisms is implemented and verified. The findings revealed the following. In external dynamic growth mechanisms, key opinion consumer (KOC) connection can lead to positive collective overreaction (i.e., the adoption rate of consumer groups spikes). This effect fades as the probability of KOC connection increases and stabilizes as the node change rate decreases. Random connection is prone to negative collective overreaction (i.e., a sudden and sharp decline in the adoption rate of consumer groups), and key opinion leader (KOL) connection exhibits both positive and negative collective overreaction. Increasing the edge change rate increases the frequency of negative collective overreaction in KOL connections. In internal dynamic growth mechanisms, KOL and KOC connections are prone to negative collective overreaction; increasing the edge change rate can reduce the frequency of negative collective overreaction in KOL overreaction, and an appropriate edge change rate can inhibit the emergence of negative collective overreaction in KOC connection. This research contributes to the area of internet product marketing and provides a new basic framework through which to combine psychology and the ABM.

1. Introduction

As e-commerce has evolved into the primary channel for product distribution, the phenomenon of collective overreaction, including behaviors such as nondurable goods rush, buying sprees, and hoarding, has emerged. This phenomenon, especially during extreme events such as the widespread panic buying of toilet paper in Japan and Australia during the COVID-19 pandemic [1], is a manifestation of individual irrationality amplified by social influence [2]. The reason lies in people, upon witnessing others making frantic purchases, overreacting to perceived scarcity of resources. Even if their own commodity reserves are sufficient, individuals may involuntarily develop a desire to make purchases. Furthermore, individual irrational behavior has an amplifying effect on social networks. Even those who are initially without a sense of crisis can be infected by social influence, ultimately leading to collective overreactions. The dynamics of online product diffusion networks significantly contribute to shaping social influence and, consequently, to the emergence of collective overreaction [3]. As a consequence, collective overconfidence may lead to unpredictable fluctuations in the evolution of online product diffusion [4]. Therefore, understanding this collective irrational behavior in the marketing process is crucial for enterprises in fortifying their brand and marketing strategies to gain a global competitive advantage [5].

In this study, collective overreaction is delineated as a phenomenon wherein a group of individuals, swayed by external influences, either embraces or relinquishes a product in a manner disproportionate to its inherent value or quality. It is stratified as positive and negative collective overreaction. The former denotes a sudden upsurge in product adoption, potentially influenced by key opinion leaders or prevailing social trends. This surge surpasses the anticipated levels based on the intrinsic value of the product. Conversely, the latter involves a precipitous and pronounced decline in product adoption, often triggered by

* Corrrsponding author at: Building Jian 4#, Gongda Road, Hongshan District, Wuhan University of Technology, Wuhan 430070, China.
E-mail address: weixiaochaowin@163.com (X. Wei).

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negative rumors, misinformation, or external influences. This leads individuals to avoid the product to a degree exceeding rational expectations grounded in its actual attributes.

As a long-standing feature of human life, overreaction cannot be explained by economic theories that rely on the assumption of rationality and have significant implications for the success or failure of new products in the market [6]. Despite the momentum of this topic, only a limited number of studies have examined the evolution of overreaction at the group level. In contrast, irrational behaviors at the individual level, including overconfidence [7], attribution bias [8], anchoring bias [9], and emotional bias [10], have been the subject of most related research. However, there are at least two fundamental limitations. First, rather than examining overreaction at the group level, these studies focused on demonstrating the existence of irrational behavior and identifying its factors at the individual level. Second, these works cannot capture the evolution of collective irrationality emerging from individual behavioral interaction in dynamic online product diffusion networks. Therefore, this research gap is bound to become a key future research direction in the field of marketing.

Compared with individual-level irrationality, due to the social influence (i.e., amplification effect, social identity, and emotional contagion) of social network dynamics, collective overreaction can effectively promote market performance. First, in online product diffusion networks, particularly for game apps, influencer marketing can indirectly amplify the purchase intentions of nonviewers through brand visibility and awareness [11]. This situation can create a feedback loop in networks where individuals respond to each other's reactions, resulting in collective overreaction that is greater than any individual response. Second, social networks contribute significantly to the establishment of social identity, prompting individuals to align their viewpoints and behaviors with the prevailing majority [12]. This social identity, in turn, can precipitate group decision-making, even when such decisions lack complete rationality. In the market, this may manifest as overbuying or overselling, driven by consensus rather than fundamental changes. Third, live-streaming services in online product diffusion networks allow for real-time concurrent consumption and commenting on media events with others, making people tend to be happier and more excited than if they were alone [13]. However, it is crucial to note that the social influence of social network dynamics can also amplify the negative repercussions of collective overreaction on market performance.

Guided by the above practical and theoretical motivations, the difficulties of this research are concentrated mainly in the following two aspects: how to describe the mechanism of collective overreaction emerging from individual irrational behaviors under the influence of social influence and how to capture the influence of the dynamics of online product diffusion networks on the collective irrationality of consumers.

To better address the shortcomings and difficulties of existing studies, we attempt to model the collective overreaction phenomenon in dynamic online product diffusion networks, which can promote the understanding of complex consumer group behavior. However, modeling the impact of dynamic social networks on consumer irrationality at the micro level is a challenge. Existing studies have pointed out that network structure can explain the occurrence of an overreaction [3]; thus, we introduce the core idea of overreaction theory to capture the influence of the dynamics of social networks on individual irrational factors. Therefore, we synthesize a complex network model, evolutionary game theory, and overreaction theory to construct the decision-making process of irrational consumers in a dynamic network environment. First, an evolutionary game model is constructed by designing a learning algorithm that utilizes overreaction theory to describe interactions among consumers. Overreaction is influenced by real-time network structure characteristics. Second, a variety of dynamic network growth mechanisms are extracted from emerging marketing strategies, and real-time network structural characteristics are obtained. Finally, a simulation model based on the ABM is established under different network growth mechanisms.

2. Literature review

2.1. Overreaction theory and collective overreaction in product diffusion

Overreaction in product diffusion is defined as people having a violent reaction to a certain piece of information that is beyond the normal reaction, such as an excessive rise in stock prices due to positive information. Overreaction is inspired by biases in the intuitive prediction of investors based on past and publicly available information [14]. Scholars have analyzed the causes of overreaction from the perspectives of psychological cognitive bias, information diffusion, and the herd effect. From the perspective of psychological cognitive bias, the Barberis et al. [15] and Daniel et al. [16] models explain how overreaction leads to price deviation from the efficient market hypothesis by using selectivity and attribution biases, respectively. From the perspective of information diffusion, the study of Hong and Stein [17] is the most representative. The above authors posited that there are two classes of investors in the market, news watchers, who trade based on information and momentum traders whose decisions depend solely on price, and who attribute overreaction to the gradual diffusion of information. The herding perspective holds that after informed traders make a decision, uninformed traders are likely to use it as a reliable signal of market price and trade. The bias of uninformed traders, such as herd behavior, causes them to overreact to signals [18].

At present, overreaction theory has been generally accepted in the research on the financial securities market. Numerous researchers have studied overreaction behavior in the global stock market [19]. A few studies have analyzed the overreaction hypothesis in other markets, such as the foreign exchange market [20], housing market [21], and Bitcoin market [22]. Because public and private information during the process of dynamic online product diffusion is similar to that in financial markets, this study adopts the core idea of overreaction theory to account for the effect of the irrational factor of the individual on internet product diffusion.

Collective irrationality has existed since the beginning of social life [23]. Theories of consumer behavior have focused mostly on the collective behavior of individual decision makers who make economic choices in relatively isolated social settings rather than on collective behavior in general, such as overconfidence [7], attribution bias [8], anchoring bias [9], and emotional bias [10]. Classical product diffusion studies, represented by the Bass model, focus on predicting macrolevel diffusion outcomes from a rational perspective but fail to gain insights into the collective irrational phenomenon among consumers [24]. Thus, the phenomenon of collective irrationality is clearly beyond the explanatory power of economic theories that rely on the assumption of rationality [6]. However, compared with studies on overreaction at the individual level, few studies have focused on overreaction behavior at the group level. Therefore, this paper attempts to analyze the evolutionary mechanism of collective overreaction emerging from individual irrationality, which is influenced by the social influences generated by the dynamics of social networks. Specifically, we attempt to incorporate the impact of dynamic network structure evolution on overreaction into a micro-adoption model, import the achievements of psychology and behavioral finance into individual decisions, and propose the modeling and simulation of collective irrational behavior in dynamic online product diffusion networks.

2.2. Product diffusion based on a complex network model

A complex network model can depict the network structure of the microcosmic world and simulate the complexity of interactions among individuals. In the early stage, Conley and Udry [25], Iyengar [26], Manchanda [27], Zhao et al. [28] and other scholars demonstrated the important influence of social network environment factors on the diffusion of new product innovation from different perspectives. In recent years, several scholars have constructed complex network models from different perspectives to study product diffusion in online social networks. Shao and Hu analyzed the diffusion of two product types using an advanced selling strategy from a social network perspective [29]. Moreover, Encarnação et al. developed a theoretical model grounded in the strategic interactions between players from governments, companies, and consumers by resorting to evolutionary game theory [30]. Huo et al. constructed a mixed node-level information diffusion model to describe the dynamic product diffusion process [31]. These studies show that complex network theory can accurately characterize the behavioral evolution of participants in product diffusion.

However, these studies are limited to static networks and cannot accurately explain the practical problems of dynamic online product diffusion in the real world. The dynamic nature of social networks cannot ignore the irrational influence of consumers. Therefore, this paper attempts to expand the complex network model, transform marketing strategies into dynamic network growth mechanisms, and simulate the impact of social network dynamics on collective overreaction.

2.3. Product diffusion based on the ABM and game theory

As one of the most widely used simulation techniques in the field of product diffusion, the ABM can be used to discover the macro-evolution mechanism of the social economic system emerging from individual micro-interaction as a bottom-up method based on interaction rules [32]. Many researchers have applied ABM to simulate viral marketing strategies on Twitter [33], social interaction in the electronic market [34], the diffusion of technological innovation behaviors [24], adoption and diffusion mechanisms of co-innovations [35], etc. Additionally, a limited number of studies have employed ABM to investigate collective overreaction in social networks. For example, Lim and Bentley utilized it to examine the extreme polarization of opinions on social media and found that it may be caused by individuals simply amplifying their own views [36]. Cinelli et al. explored the key differences between the main social media platforms and how they are likely to influence information spreading and echo chambers' formation [37]. However, the interaction rules set by models in existing ABM studies are relatively simple and unconvincing. Therefore, it is difficult to accurately describe individual interactions in online product diffusion networks.

Game theory, as one of the main frameworks for behavioral decision analysis, is characterized by simplicity, efficiency, and strong analytical ability [38] and can be used to set individual interaction rules to compensate for the defects of the ABM. At present, a few scholars have combined ABM and game theory to study individual interaction behavior [39]. Therefore, this paper introduces it into the ABM as the minimum component unit of individual decision making and designs interindividual learning imitation rules to describe how individual interactions change decision making.

3. Research framework and conceptual model

3.1. Research framework

The collective overreaction of dynamic online product diffusion stems from the dynamics of the social network structure (see Fig. 1). This phenomenon emerges from individuals' decisions to buy or reject products in combination with various information and psychological factors in complex and dynamic online product diffusion networks. Marketing strategies change the network structure of online product diffusion networks, which affects individuals' utility judgments and leads to decision-making bias (i.e., overreaction). Individual irrationality is amplified by online product diffusion networks, and eventually, unpredictable collective irrationality emerges.

This paper adopts the ABM to analyze collective overreaction in dynamic online product diffusion networks. Several well-established theories, including complex network theory, evolutionary game theory, and overreaction theory, are employed to construct the research framework. This research framework is presented in Fig. 2. Step 1: By expanding the innovation diffusion process to overcome static network

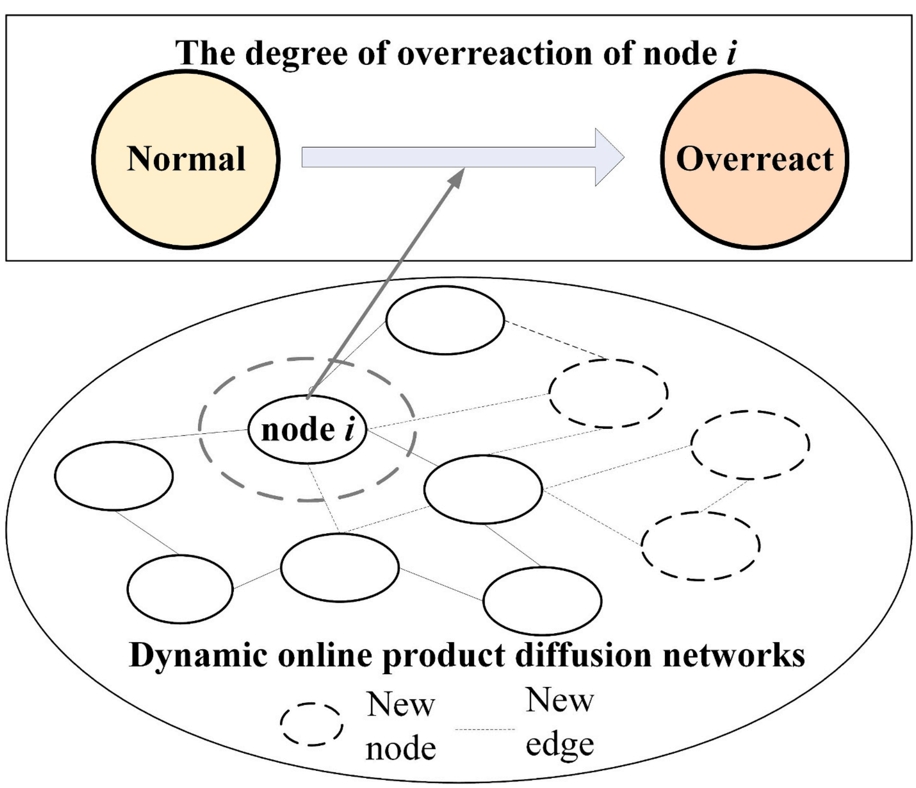

Fig. 1. The dynamic online product diffusion networks influence individual overreaction.

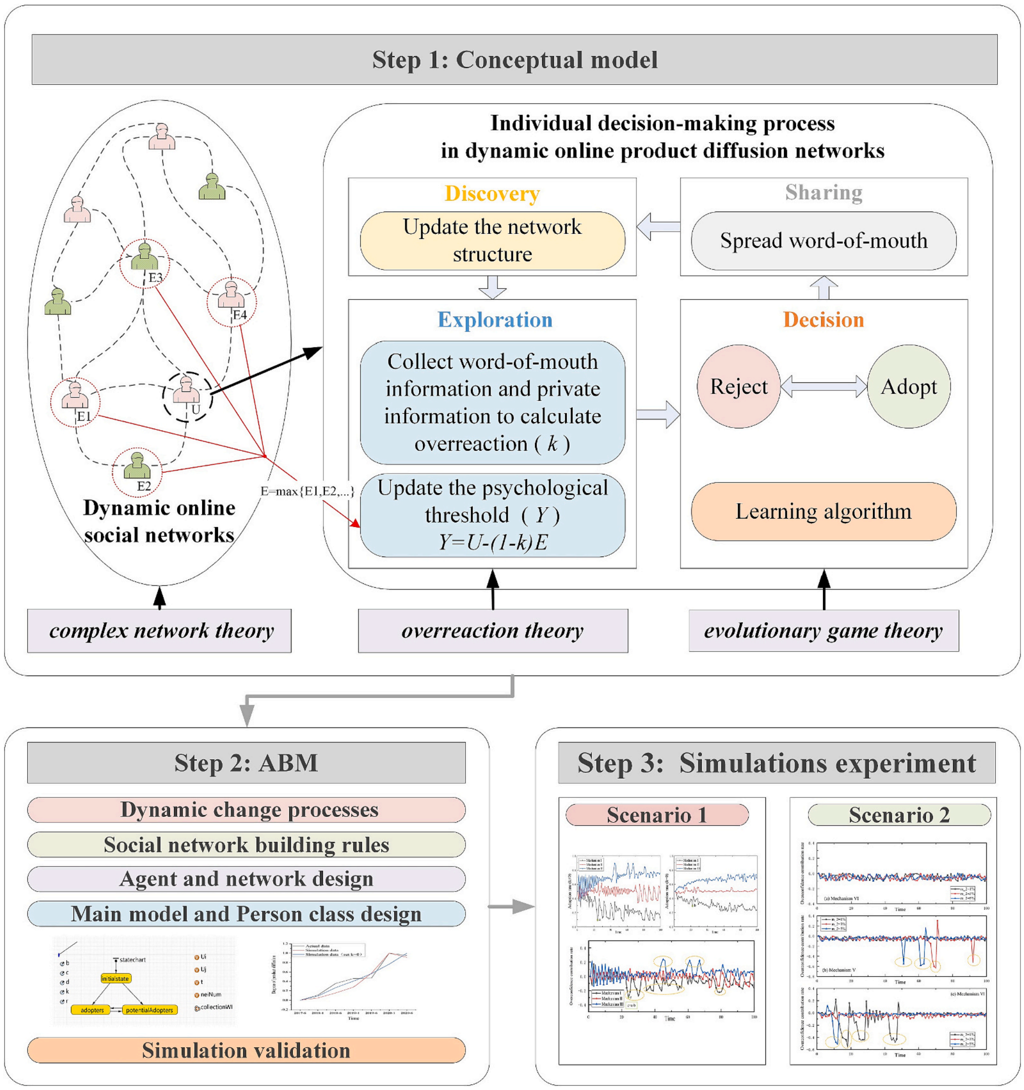

Fig. 2. Research framework.

constraints [40], a four-stage conceptual model is established to describe the dynamic product diffusion process. An evolutionary game model that considers the effects of overreaction is built as the underlying logic of individual decision making. Step 2: To elucidate how various marketing strategies affect the dynamic evolution of network structure, we design six dynamic growth mechanisms of networks based on complex network theory. This design is informed by the core concepts of key opinion leader (KOL) and key opinion consumer (KOC) promotion, as well as insights from prominent academic models. Build an ABM that describes group behavior evolution with the evolutionary game from Step 1. It has been verified that our proposed model reflects dynamic online product diffusion in the real world better than does a model that ignores overreaction. Step 3: Simulate the evolution of collective overreaction in dynamic online product diffusion networks under various network dynamic growth mechanisms and collect and analyze the simulation results.

3.2. Conceptual model

Dynamic online product diffusion is a process in which bounded rational individuals constantly learn from each other and evolve under the influence of private and public information, irrational factors, and network structure changes. Marketing strategies cause changes in the network structure, which leads to subsequent changes in private and public information and irrational factors. In particular, overreaction can bias individuals' utility perceptions and make group behavior unpredictable.

The innovation decision process serves as the foundational structure within the conceptual framework [40], encompassing five pivotal stages: knowledge, persuasion, decision, implementation, and confirmation. However, conventional applications of this process often rely on static network representations, neglecting the nuanced dynamics inherent in real-world social networks. To address this limitation, our study extends it to a four-step dynamic online product diffusion model, illustrated in step 1 of Fig. 2. The discovery phase involves the social network actively engaging new consumers or potential adopters, fostering connections among them within the network. This interaction may take the form of users following KOLs on platforms such as Facebook. As new consumers are integrated and fresh connections emerge among them, a discernible transformation occurs in the structure of the consumer network. This distinction serves as a crucial determinant between the dynamics of network diffusion and the more inert characteristics of static network diffusion. Progressing to the exploration stage, potential adopters engage in informational interactions, dynamically adjusting psychological thresholds influenced by overreaction phenomena [41]. The decision stage involves potential adopters evaluating the adoption of products based on individual decision rules. Finally, in the sharing stage, adopters assess expected versus actual revenue, disseminating word-of-mouth (WOM) information throughout the evolving network structure.

The following assumptions are made for the research subject. A product is a general product sold on social networks without special attributes [38]. There is only one product in the market without any substitute [38]. A product has a positive network externality, i.e., non-purchasers can benefit from others' purchases but with a small revenue loss [42]. Users are boundedly rational and can obtain decision and WOM information only from their neighbors [43].

3.2.1. Overreaction under the influence of a dynamic network

The core ideas of overreaction theory include the following: individuals tend to imitate others, overweight immediate information and underweight past information [15,18], and with the gradual diffusion of public information, individuals increasingly trust private information when it is consistent with public information [16,17]. A similar behavioral logic exists in dynamic online product diffusion: when the number of users using a product increases, the utility that each user obtains from using the product also increases, and vice versa. However, in social networks formed by different evolutionary mechanisms, the degree of consumer overreaction often varies. Thus, it is important to understand how different factors influence consumer overreaction in dynamic online product diffusion.

We consider the factors influencing the degree of consumer overreaction at the micro and macro levels. At the micro level, according to the core idea of overreaction theory, we consider the effect of short-term WOM information (α) on consumers' overreaction behavior. Assume that individuals consider only the current time t and the previous time $t-1$ of the WOM information. The values of short-term WOM information are shown in Table 1. At the macro level, many studies attribute this overreaction to excessive network centrality [3]. Real-time changes in social network topology cause changes in network centrality. The degree centrality is the most direct measure used to describe node centrality and can be expressed as $C_{deg}(i) = \frac{d_i}{n-1}$, where n is the total number of nodes to which node i belongs and d_i is the degree of node i. The average centrality of this network at time t is $\bar{C}_t = \frac{1}{n}\sum_{i=1}^{n} C_{deg}(i)$.

Thus, when the individual's decision is consistent with public information (α), that is, when the individual holds the adoption strategy and $\alpha = 1$ or holds the rejection strategy and $\alpha = -1$, they will feel numb to changing their strategy. We defined the degree of numbness as k. However, when individuals' decisions are inconsistent with public information, they are more willing to change their strategies. We also define the degree of sensitivity as k, which can be represented as $k = \alpha^*\bar{C}_t$. A larger k indicates a greater probability of deviation from the normal situation.

3.2.2. Modeling the product diffusion process

Individuals' decision-making behavior during dynamic online product diffusion can be described by an evolutionary game model (as shown in Table 2). Individuals may have two strategies—adoption or rejection—indicating that consumers choose to buy (or use) the product or reject it, respectively. The adoption of products brings about benefits (b) for the user, such as entertainment and satisfaction. Such adoption also requires the user to pay a cost (c), such as time or money, which is shared among all adopting individuals. Because of the positive network externality, consumers who reject the product also enjoy these benefits, but there is some loss of revenue, expressed as a penalty (b), which is shared by all rejecting individuals. n denotes the total number of individuals, and x is the proportion of adopters. In pursuit of maximizing desired utility, consumers present game behavior by considering a combination of benefits, costs, penalties, and the strategies and benefits of their neighbors and by switching between adoption and rejection decisions.

Table 1
The value of short-term WOM information.

The WOM condition of time t	The WOM condition of time $t-1$	
	More positive WOM	More negative WOM
More positive WOM	$\alpha = 1$	$\alpha = 0$
More negative WOM	$\alpha = 0$	$\alpha = -1$

Table 2
The interactive behavioral game payoff matrix in product diffusion.

Player 1	Player 2	
	Adoption	Rejection
Adoption	$b-c/nx, b-c/nx$	$b-c/nx, b-d/[n(1-x)]$
Rejection	$b-d/[n(1-x)], b-c/nx$	$0, 0$

The expected utility of adoption can be expressed as $E_a = b - \frac{c}{nx}$. The expected utility of rejection is $E_r = x\left[b - \frac{d}{n(1-x)}\right]$. The average expected utility of this group is $\bar{E} = xE_a + (1-x)E_r$. The game equilibrium equation is $bnx^3 - (2bn-d)x^2 + (bn+c)x - c = 0$. According to evolutionary game theory, homogenous players develop imitative behavior with equal probability.

However, the change in network topology and the overreaction of consumers make it difficult to obtain an equilibrium point in this game model because they would lead to the perceived bias of neighbors' expected utility. Therefore, we introduce overreaction (k) to fully consider the impact of irrational behavior on product diffusion. The main idea of overreaction theory can be represented by the fact that individuals develop a perceptual bias toward the utility of neighbors with nonidentical strategies, which causes individuals to change their probability of imitating their neighbors. Therefore, the expected utility of adoption for rejecting individuals can be expressed as $E_a = (1+k)\left(b - \frac{c}{nx}\right)$. Similarly, the expected utility of rejection for an adopting individual is $E_r = (1-k)x\left[b - \frac{d}{n(1-x)}\right]$.

The evolutionary game process under network evolution in a dynamic network environment is shown in Fig. 3. According to the conceptual model shown in Fig. 2, at each time t, micro-individuals go through the following four processes in making the decision to adopt a product:

1) **Discovery.** The network structure is updated to Λ_t under the current time t according to different network growth mechanisms. Due to the addition of new nodes or the establishment of new edges, the status of each node's neighbors and the degree of network centrality may also change.
2) **Exploration.** Individuals first collect the WOM of their neighbors, update WI, and then update the degree of overreaction k. The psychological threshold function of products adopted by individual i is expressed as $Y = U_i - (1-k)^*max(E_j)$, where U_i is the utility of i and j is a neighbor of i.
3) **Decision.** Game evolutionary theory states that every participant adjusts their strategies to improve their revenue [44], so individuals follow and learn the strategy based on the probabilistic model when $Y < 0$. The probability of individuals changing their strategies can be expressed as $p_{(b_i \to b_j)} = \frac{1}{1+e^{\frac{Y}{r}}}$, where r represents information noise and $p_{(b_i \to b_j)}$ decreases as the value of r increases. At this stage, individuals maintain the original strategy or transform it into a new strategy, and the environmental information in the next period ($t+1$) also changes.
4) **Sharing.** The adopter first calculates his or her actual utility. The utility of the product adopter at time t can be expressed as $U_a(\Lambda_t) = b - \frac{c}{n_t x}$. Second, the adopter compares the expected utility with the actual utility to obtain WOM ($W = Sgn[U_a(\Lambda_t) - E_a(\Lambda_{t-1})]$) and spreads it to his or her neighbors.

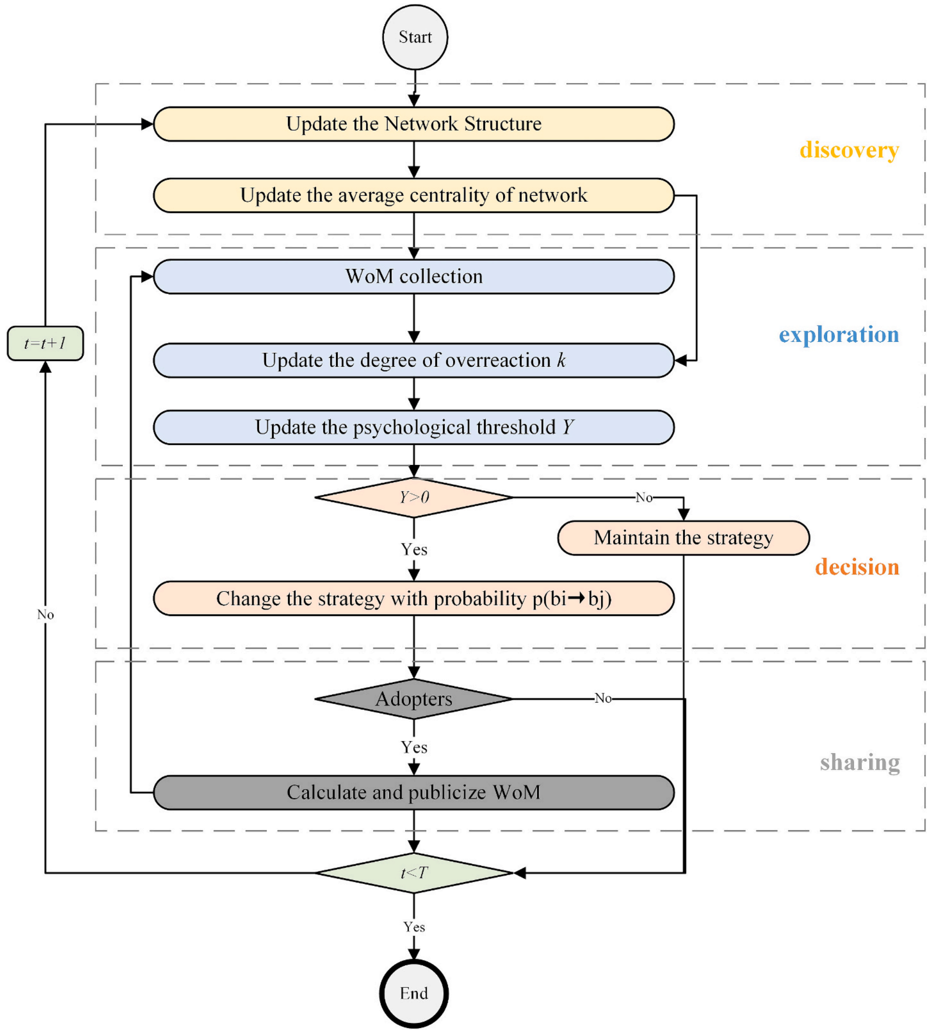

Fig. 3. Flowchart of the model.

4. ABM design and validation

The ABM is a well-established methodology for addressing complex systems [45], and its design starts with the design of the agent and network. An agent represents an individual or organization in a model. Given that product diffusion pertains to actual users, we use agents to represent users, while the agents and their connections constitute the product diffusion network. Subsequently, the main model, person class, and experimental system are designed. After various validations and adjustments to the model, the conceptual model is implemented in the multi-agent model.

4.1. Dynamic change processes of diffusion networks

Social networks in reality display attributes of small-world networks [46]. To better reflect the reality of online product diffusion networks, we first generate a special small-world network as the initial hypernetwork leveraging the characteristics of the Facebook dataset (refer to Table 3). An online social network can be regarded as consisting of nodes (users) and their connected edges (relationships among users) with each other. New potential adopters are constantly attracted and connected with other nodes (such as KOLs and KOCs) by marketing strategies. This situation changes the topology of the online social network, making the network not simple and static but complex and dynamic. KOL and KOC promotion are two representative types of marketing strategies. A KOL refers to a node that has greater influence and discourse power in an online social network, while a KOC is a node that can influence the consumer behavior of its followers. Compared with a KOL, a KOC has fewer followers and less influence but is more direct and less expensive.

Table 3
Dataset summary for Facebook.

Name of Index	Value
Number of nodes	63,731
Number of edges	1,634,180
Average degree	25.6
Average clustering coefficient	0.253
Number of triangles	3,501,542

Marketing strategies can be transformed into a variety of network growth mechanisms. According to the core points of KOL and KOC promotion and mainstream academic research, we divide the ways in which new connecting edges are generated into three types: random connection, KOL connection, and KOC connection. Depending on whether new nodes are added, dynamic network growth mechanisms can be divided into two types: external dynamic growth mechanisms (i.e., adding new nodes) and internal dynamic growth mechanisms (i.e., not adding new nodes). Thus, six network dynamic growth mechanisms can be formed.

1) **External dynamic growth mechanisms**

External dynamic growth mechanisms refer to network evolution due to new nodes joining and connecting to old nodes. These mechanisms represent the normal state of network evolution. For example, WOM communication, advertising, etc., may turn unaware users into potential users.

Mechanism I: External growth, random connection. It means that a node is connected to any node with the same probability. Substep 1: At each time t, z new nodes are added to the network. Substep 2: Each new node is connected to the original nodes in a completely random way until the number of edges reaches m_1. Substep 3: Repeat substeps 2–3 until $t = T$ and network evolution are complete.

Mechanism II: External growth, KOL connection. This means that new network nodes are more likely to connect with original nodes that have a higher degree value (i.e., number of neighbors), coinciding with the core of the BA model [47], which is an influential network evolution model and simulates KOL promotion in online product diffusion networks. For example, on social media sites, users with more followers gain followers faster and more easily than do those with fewer followers. Substep 1: At each time t, z new nodes are added to the network. Substep 2: Each new node is connected to the original nodes with probability p until the number of edges reaches m_1. The calculation of p is expressed as $p = \frac{d_i}{\sum_{j=1}^{n} d_j}$, where d_i is the degree of node i, which is the number of neighbors of node i, and n is the total number of nodes. Substep 3: Repeat substeps 2–3 until $t = T$.

Mechanism III: External growth, KOC connection. The core mechanism is that when new nodes tend to connect with neighbors of connected nodes, it reflects the core mechanism of KOC promotion in online product diffusion networks and the HK model [48], which uses the influence of KOC to connect with neighbors' neighbors. For example, user A's retweeting of user B's tweets may lead to user A's followers following user B. Substep 1: At each time t, z new nodes are added to the network. Substep 2: Each new node is first connected to one node i in the KOL connection mechanism. Substep 3: Select a node, j, randomly from the neighbors of node i to connect with probability P_{KOC}. If the connection fails, then the KOL connection is performed again. Substep 4: Repeat substep 3 until the number of edges reaches m_1. Substep 5: Repeat substeps 2–4 until $t = T$.

2) **Internal dynamic growth mechanisms**

Internal dynamic growth mechanisms are also key factors for product proliferation, as they are considered network evolution caused by adding edges between original nodes while maintaining the same number of nodes. For example, some products may lack channels through which to attract new users because their target users are too niche.

Mechanism IV: Internal growth, random connection. Substep 1: At each time t, two unconnected nodes are randomly selected, and an edge is added. Substep 2: Repeat substep 1 until the number of new edges reaches m_2. Substep 3: Repeat substeps 1–2 until $t = T$.

Mechanism V: Internal growth, KOL connection. Substep 1: At each time t, randomly select one node i. Substep 2: Node i is connected to other nodes with probability p. Substep 3: Repeat substeps 1–2 until the number of new edges reaches m_2. Substep 4: Repeat substeps 1–3 until $t = T$.

Mechanism VI: Internal growth, KOC connection. Substep 1: At each time t, randomly select one node i. The neighbor of node i is denoted as $N_1(t)$, and the neighbor of node $N_1(t)$ is denoted as $N_2(t)$. Substep 2: Randomly select one node j in $N_2(t)$ and connect it to node i. Substep 3: Repeat substeps 1–2 until the number of new edges reaches m_2. Substep 4: Repeat substeps 1–3 until $t = T$.

4.2. The Main model and the person class design

According to the conceptual model in step 1 of Fig. 2, the settings of the main model and the Person class are shown in Fig. 4-5Figure 5. The main model is mainly used for setting up the initial network and network evolution rules, collecting and presenting data, etc. The person class represents the individual behavior logic of the agent and contains parameters, variables, and state transformation diagrams. (See Fig. 5.)

4.3. Simulation validation

ABM validation is a complex task due to the inclusion of numerous parameters and assumptions [49]. Thus, we performed conceptual validation, internal validation, micro validation, and macro validation. First, we performed a conceptual validation by verifying that the related theories were appropriately integrated into our model. The results show that their combination is helpful for describing the collective overreaction mechanism emerging from individual irrational behaviors, in which complex networks, evolutionary game theory, and ABM are suitable for describing the dynamic evolution of networks, the interaction rules between individuals, and the macro emergence of individual interactions, respectively. Second, we altered the values of the input variables and parameters to extreme values to test whether the model worked as expected. For instance, when information noise is minimal, the level of product diffusion fluctuates less over time, whereas the opposite is true when information noise is significant. This finding corresponds with existing research showing that the greater the noise in network information is, the more unstable users' behavior becomes [50]. Third, micro validation focused on the dynamics of the network. Specifically, each mechanism was carefully calibrated and tested to ensure that the simulated network dynamics resembled those observed in real-world scenarios. By observing the visualized dynamic evolution of networks, it was found that all six network dynamic growth mechanisms were successfully implemented. By observing the output network metrics such as the degree distribution, clustering coefficient, and small-world properties, it was concluded that the networks generated by our model exhibit characteristics consistent with real-world networks.

Fourth, the proposed simulation model was macro-validated using real market data from China. We used TikTok as a case study because it has a representative user base in terms of product dissemination and collected data on average daily active users in China from June 2017 to June 2020. Then, we adjusted the parameters according to the real situation and reran our simulation model 1000 times to obtain the average simulation results. By comparing the normalized actual data with the simulation data, it can be seen that the trends of the three lines are roughly the same, and the accuracy of the $k = 0$ case is slightly worse (as illustrated in Fig. 6). This is consistent with Rand and Rust's guidelines for rigor in ABM; that is, real-world data validation involves showing that the average model results correlate with the real-world results [51]. The validation results indicate that, to some extent, our model accurately captures real-world behavior.

5. Simulation study

To investigate the impact of network dynamic growth mechanisms on the evolution of collective overreaction in dynamic online product

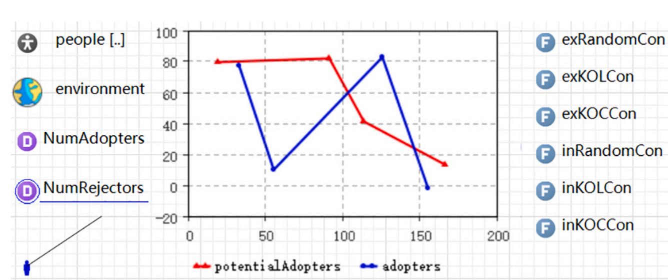

Fig. 4. Main model description.

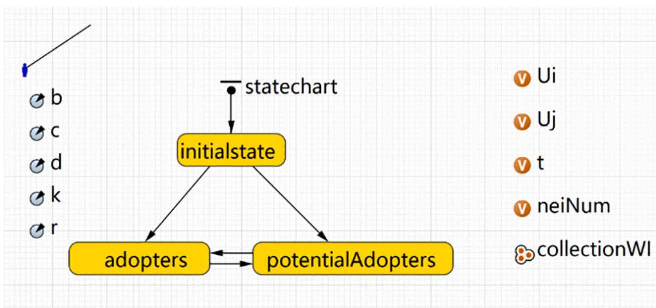

Fig. 5. Person class description.

diffusion networks, we conducted a series of simulation experiments. The initial parameter values of the model were determined based on actual market conditions or existing research, and we modified one variable at a time while keeping the other variables constant. The product utility parameters were held constant while we analyzed the evolution of consumers' decision making over time. To ensure the accuracy of the simulation results, we repeated each group of experiments 1000 times to obtain the average simulation results. The quantitative results of the final results of 1000 runs are reported in the Appendix A, and ANOVA was employed to test the differences in means among multiple sets of experiments at the 95% confidence level.

5.1. Default parameters

Table 4 provides an illustrative example of the parameter settings for external dynamic growth mechanisms. To better reflect real-world conditions, we assumed that each agent had an equal likelihood of adopting or rejecting the product at the outset, resulting in a 50% initial adoption rate. According to the work of Du et al. [52], the information noise was set at $r = 0.01$. We further assumed that no additional edges were added to the initial node in this dynamic growth mechanism ($m_2 = 0$). To better understand the dynamic evolution of the network, we also introduce two additional parameters, R_{ni} and R_{ed}, which represent the inactivity rate of nodes and the disconnection rate of edges, respectively, both initialized to 0. By varying the values of parameters n, m_1, and P_{KOC}, we investigated the impact of network dynamic growth mechanisms (mechanisms I-III) on diffusion outcomes. For internal dynamic growth mechanisms, the parameter settings were manually adjusted during each simulation to reflect the unique circumstances.

5.2. External dynamic growth mechanisms

5.2.1. Impact of network dynamic growth mechanisms

To better visualize the evolution of collective overreaction, we utilized the overreaction contribution rate (OCR) as a performance evaluation indicator. Specifically, we set $k \neq 0$ to denote overreaction and $k = 0$ to denote not considering overreaction. We then ran the simulation model to obtain the adoption rates in both cases. Finally, we compared the adoption rates at each interval in both scenarios to

Table 4
Default settings of the parameters in the experimental system.

No.	Parameter	Description	Default
1	Network-type	Type of network	Small-world
2	P-adoption	Probability of initial nodes holding adoption strategy	0.5
3	r	Information noise	0.01
4	z	Node change rate: the rate of the number of new nodes to the number of original nodes in each time step	3%
5	m_1	Edge change number for new node: number of new edges added for each new node at each time step	3
6	m_2	Edge change rate: the number of new edges to the number of original edges in each time step	0
7	P_{KOC}	Probability of KOC connection	0.5
8	R_{ni}	The inactivity rate of nodes	0
9	R_{ed}	The disconnection rate of edges	0

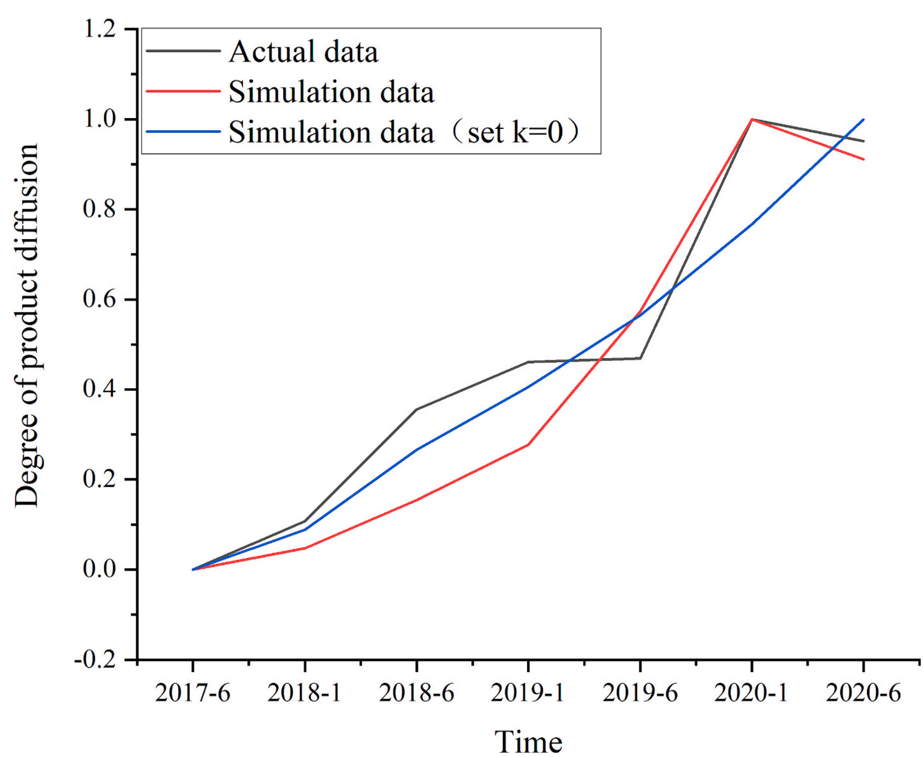

Fig. 6. Comparison of simulation outcomes with real case data.

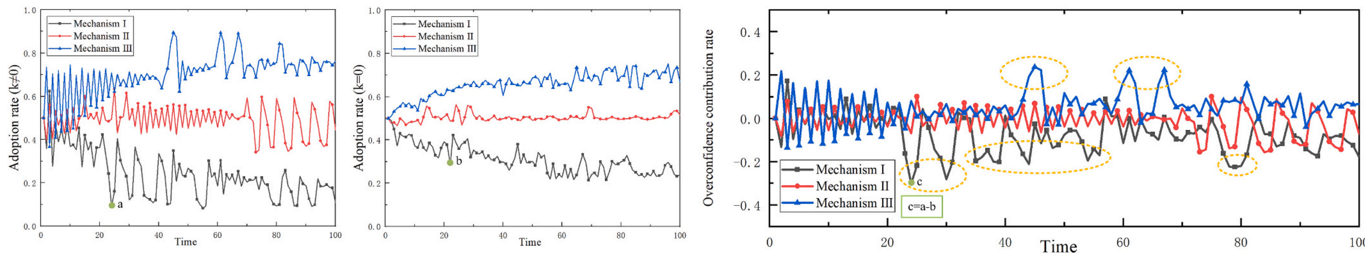

Fig. 7. Results of product diffusion under different network external growth mechanisms.

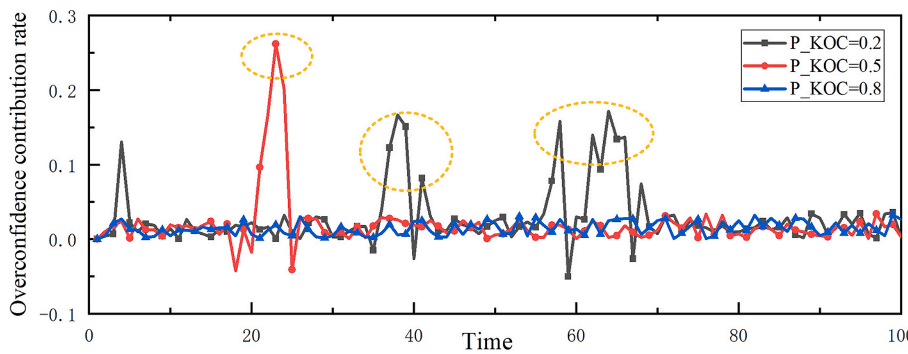

Fig. 8. Results of product diffusion under different P_{KOC}.

determine the OCR (for example, $c = a - b$, as depicted in Fig. 7).We compared three mechanisms of network evolution (I, II, and III) to observe their impact on the OCR. The results suggested that the product diffusion network generated by Mechanism III can lead to positive collective overreaction, characterized by a spike in the OCR followed by a return to normal levels (as depicted in Fig. 7). In contrast, the product diffusion network produced by Mechanism I is more likely to result in negative collective overreaction, while that produced by Mechanism II exhibits both positive and negative collective overreaction. This finding corresponds to the reality that KOC promotion strategies rely heavily on trust among familiar relatives or friends and that new potential adopters are closely linked to existing adopters. A KOC has a greater decision-making influence on vertical user masses than does a KOL or a stranger; thus, KOC promotion tends to achieve positive collective overreaction, higher sales volume, and better WOM.

5.2.2. Impact of the KOC connection probability in mechanism III

Furthermore, we set P_{KOC} to 0.2, 0.5, and 0.8 to investigate how the extent of KOC recommendation affects the OCR. Fig. 8 illustrates that various P_{KOC} values exhibit comparable overall OCRs, yet excessively high P_{KOC} values can lead to collective overreaction vanishing. This phenomenon is attributed to the fact that when too many consumers become KOCs, a low threshold undermines the quality of promotion and may even have an adverse impact. For example, some bloggers on Xiaohongshu (a Chinese social media platform similar to Instagram) share heavily edited photos of tourist attractions that disappoint users who visit those attractions after seeing the photos.

5.2.3. Impact of the node change rate

To examine the impact of the node change rate on the OCR, we set z to 1% and 5%. As z increases, the collective overreaction phenomenon disappears more quickly, which makes the product adoption rate converge faster, especially in Mechanism II (Fig. 9). This situation implies that increasing the node change rate is conducive to promoting the market to reach stability sooner. Consistent with the example of epidemic transmission, when governments and public health institutions implement more frequent and timely control measures, epidemics may be controlled and stabilized more quickly. An increase in the node change rate indicates that the number of people exposed to the promotion strategy increases. Thus, enterprises should choose KOLs related to their target group's vertical field for a better market when conducting KOL marketing.

5.2.4. Impact of the number of edges added for each new node at each time

To observe the effect of the number of edge changes on the OCR, we set m_1 to 2, 5, and 8. With the increase in m_1, negative collective overreaction phenomena occur more frequently in the product diffusion network derived from Mechanism II (Fig. 10 a), but the results for Mechanism III do not vary much (Fig. 10 b). This finding is consistent with reality. For example, Jiaqi (Austin) Li, one of the most popular KOLs in the Chinese beauty industry, caused consumers to boycott the beauty brand Florasis, with which he was closely associated, due to his

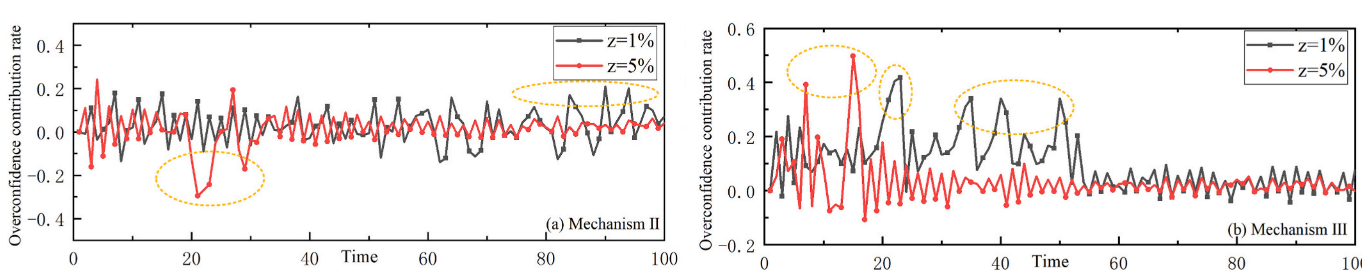

Fig. 9. Results of product diffusion under different z.

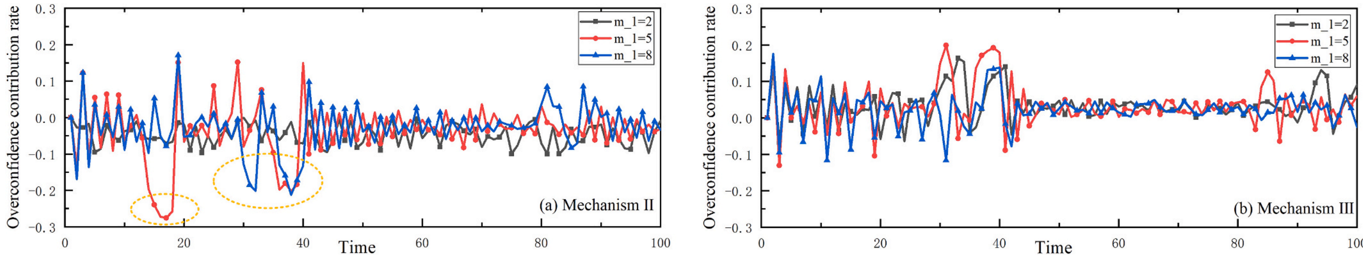

Fig. 10. Results of product diffusion under different m_1.

erroneous remarks. The reason for this is that excessive KOL promotion reduces network connectivity. If a KOL criticizes or questions the product, then it will inevitably lead some nodes to turn the strategy into rejection. It can be concluded that KOL promotion is a double-edged sword and should be used with caution.

5.2.5. Impact of the inactivity rate of nodes and the disconnection rate of edges

Furthermore, to examine the inactivity rate of nodes and the disconnection rate of edges on the OCR in mechanisms II and III, we set R_{ni} and R_{ed} to 5% and 10%, respectively. With increasing R_{ni} and R_{ed}, there was little change in the phenomenon of collective overreaction, but this had a negative effect on the level of product diffusion (Fig. 11). This finding is similar to that of previous research. Obviously, this is consistent with the real world. On social media platforms, if the number of inactive users increases or if they stop following the brand or KOL, it will hinder the spread of information in the network and reduce the diffusion level of the product.

5.3. Internal dynamic growth mechanisms

5.3.1. Impact of network dynamic growth mechanisms

We compared Mechanisms IV-VI without the addition of new nodes ($n = 0$ and $m_1 = 0$) and found that those product diffusion networks derived from Mechanisms V and VI are prone to negative collective overreaction, i.e., a sudden and sharp decline (marked in Fig. 12). This finding is due to Mechanism IV increasing network connectivity more than Mechanisms V and VI. When product promotion is in the bottleneck period, random recommendations can be adopted. For example, in the first half of 2023, when the conversion rate of KOLs and KOCs gradually declined, the beauty company *KANS* pioneered the use of short dramas for product promotion and firmly focused on conducting live broadcasts themselves instead of relying on KOLs for product promotion. As a result, they achieved sales exceeding 1 billion RMB on Douyin and topped the monthly beauty sales charts on the platform multiple times. The explanation lies in irrational consumers overreacting to the evaluation or promotion of a KOL or a KOC in the absence of new nodes, which is more likely to reverse in the future, and the phenomenon of collective irrationality will become particularly significant. Mechanism IV further increases network connectivity. Therefore, when product promotion is in a bottleneck period, enterprises should avoid the use of KOL or KOC promotion and can instead use random recommendations.

5.3.2. Impact of edge change rate

We examine how m_2 affects the OCR. The results (Fig. 13) revealed that although increasing m_2 has little effect on the product diffusion network derived from Mechanism IV, it can reduce the frequency of negative collective overreaction derived from Mechanism V. For the networks derived from Mechanism VI, an appropriate edge change rate can inhibit the emergence of negative collective overreaction. For example, China's e-commerce platforms often hold promotional events, such as Singles' Day and 618, to entice consumers to buy goods by offering discounts, coupons and limited-time specials, thereby boosting sales. When the product diffusion network is in a bottleneck period, it is necessary to choose the best marginal change rate according to the evolution mechanism of the product diffusion network to suppress the occurrence of negative collective overreaction.

5.3.3. Impact of the inactivity rate of nodes and the disconnection rate of edges

Furthermore, to examine the inactivity rate of nodes and the disconnection rate of edges on the OCR in mechanism IV, we set R_{ni} and R_{ed} to 1%, 5%, and 10%, respectively. It was counterintuitively found that the collective overreaction phenomenon changed only slightly with increasing R_{ni} and R_{ed}, but appropriate R_{ni} and R_{ed} helped to increase the level of product diffusion (Fig. 14). This is similar to some situations in reality. When the market is highly competitive and it is difficult to attract new customers, appropriately reducing marketing activities to already active customers and focusing on potential customer groups may be more effective in increasing the level of product diffusion.

6. Discussion and conclusions

This research has explored how network dynamic growth mechanisms affect the evolution of collective overreaction in dynamic online product diffusion networks. An ABM was developed to capture the dynamic interactions among consumers and products in online product diffusion networks. Our findings reveal that different network dynamic growth mechanisms have different impacts on collective overreaction, such as whether it occurs at all, whether it is positive or negative, how often it occurs and how quickly it disappears.

6.1. Summary of findings

6.1.1. External dynamic growth mechanisms

First, KOC connections can cause positive collective overreaction, but this effect fades with a high KOC connection probability. However, this random connection is prone to negative collective overreaction, and KOL connection exhibits both positive and negative collective overreactions. Network external growth evolution mechanisms significantly affect product diffusion results. New nodes connected with a KOL generate positive collective overreaction more than do random connections. Adding a KOC connection to a KOL connection increases this effect and leads to positive collective overreaction. This effect becomes more stable, and collective overreaction disappears with a higher KOC connection probability. The reason is that Mechanism II (KOL connection) makes the network more scale free, and KOLs' decisions influence other consumers more easily. KOC promotion relies on trust among familiar people, and new potential adopters are linked to existing adopters. The credibility of a KOC is greater than that of a KOL or a stranger. The simulation results match the realistic scenario of product diffusion. For example, short video platforms such as TikTok and YouTube Sign Head KOLs attract more fans. Taobao and Pinduoduo use "friends help" activities for fission marketing in China based on "KOC recommendation". Therefore, marketing activities should target communities with large nodes, use KOL promotion to draw new users and

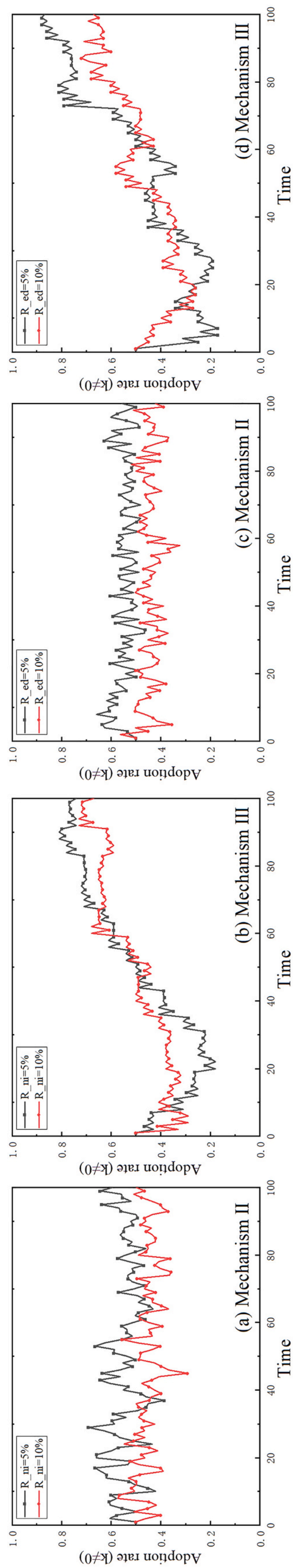

Fig. 11. Results of product diffusion under different R_{ni} and R_{ed}.

guide their purchases, and use KOC promotion at the same time, especially in online communities with large aggregation.

Second, in the product diffusion network generated by KOL recommendation, the collective overreaction phenomenon disappears faster as the node change rate increases. The node change rate represents the people reached by the advertising campaign, and thus, increasing its value will increase the number of adopters. The coverage of KOL promotion is always wider than that of KOC promotion, but the probability of success is average. Thus, it is extremely important to expand the reach and improve the reach efficiency of advertising campaigns. When conducting KOL marketing, enterprises should choose KOLs related to the vertical field of the target group to better market to the target group.

Third, with the increase in edges added for each new node at each time, negative collective overreaction phenomena occur more frequently in the product diffusion network derived from KOL promotion, but the KOC connection results do not vary much. The reason for this is that KOL promotion weakens network connectivity and increases key node influence when the number of edges added for each new node at each time is large, leading to a tit-for-tat situation between adopters and rejecters. In contrast, the KOC connection maintains high network connectivity and clustering, regardless of the number of edges added for each new node at each time, resulting in a steady increase in the number of adopters. Therefore, our results suggest that network connectivity and clustering are crucial for product diffusion. As KOL promotion can be risky if it reduces network connectivity, KOC promotion can be more effective if it leverages social relationships among existing users.

6.1.2. *Internal dynamic growth mechanisms*

First, there is a negative collective overreaction in KOL and KOC connections, and random connections can stabilize the OCR. For internal dynamic growth mechanisms, the influence of various connectivity mechanisms is smaller than that for external dynamic growth mechanisms. Among the internal growth methods of the network, random connection edges are the optimal choice for maintaining the stability of the adoption ratio. In contrast, there is a negative collective overreaction in KOL and KOC connections. The KOL connection runs the risk of falling into a lower purchase ratio, which is due to the higher level of diffusion and average gain due to the network connectivity that is further increased by the random connection edges and the more pronounced network clustering. In contrast, KOL connection makes the network significantly scale free, which results in greater fluctuations in the changes in the number of adopters and less stability in product diffusion. That is, product rejectors with large degree values in a network structure with a constant node size may have a negative impact. KOC connection helps mitigate this negative impact. Therefore, when product diffusion is at a bottleneck, random recommendations should be adopted to increase individual connectivity in the existing product diffusion network. In addition, it is necessary to focus on the key nodes that receive low benefits from the product or give the product negative reviews, and meeting the needs of this group and controlling risks are the key factors through which to promote product diffusion.

Second, although increasing the edge change rate has little effect on the product diffusion network derived from random connections, it can reduce the frequency of negative collective overreaction derived from KOL overreaction. An appropriate edge change rate can inhibit the emergence of negative collective overreaction. When the product diffusion network is in a bottleneck period, it is necessary to choose the best marginal change rate according to the evolution mechanism of the product diffusion network to suppress the occurrence of negative collective overreaction.

6.2. *Theoretical contributions*

This study contributes to the literature in three ways. First, our study contributes to the irrational behavior literature that has focused mainly on individual-level mechanisms, including overconfidence [7,53,54],

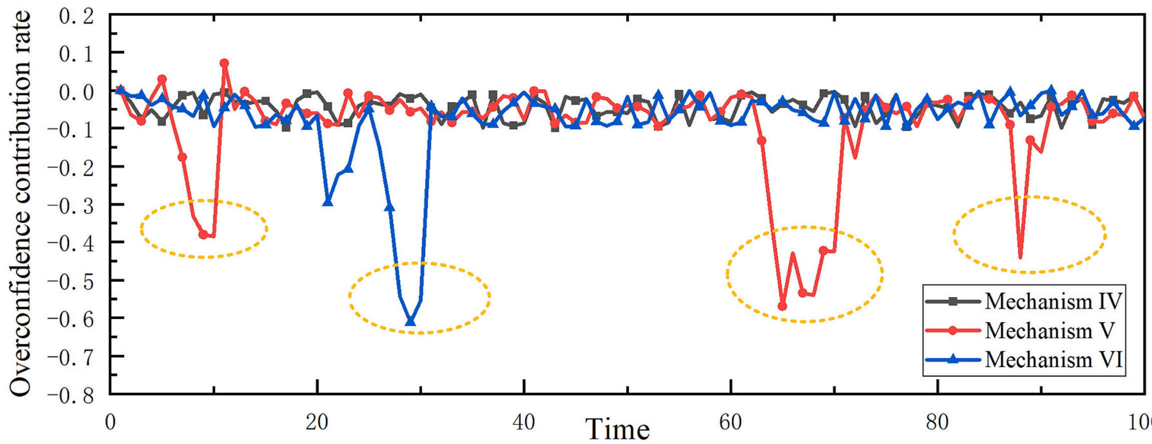

Fig. 12. Results of product diffusion under different network internal growth mechanisms.

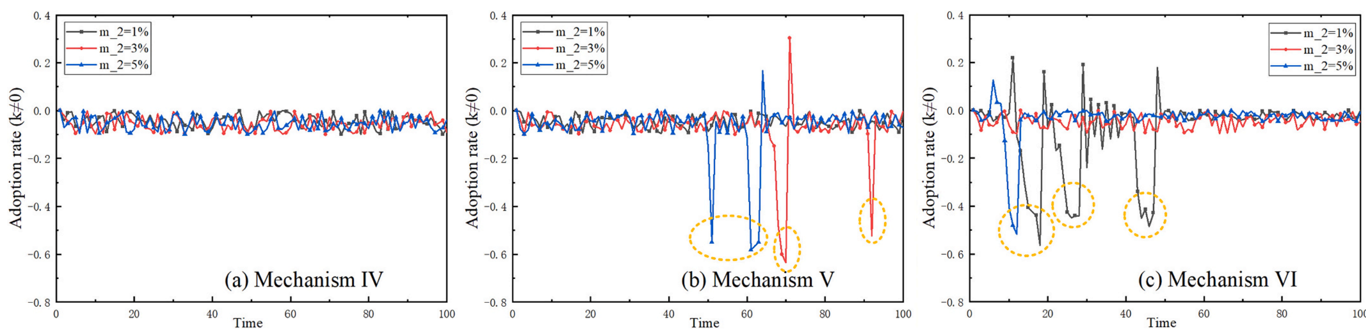

Fig. 13. Results of product diffusion under different m_2.

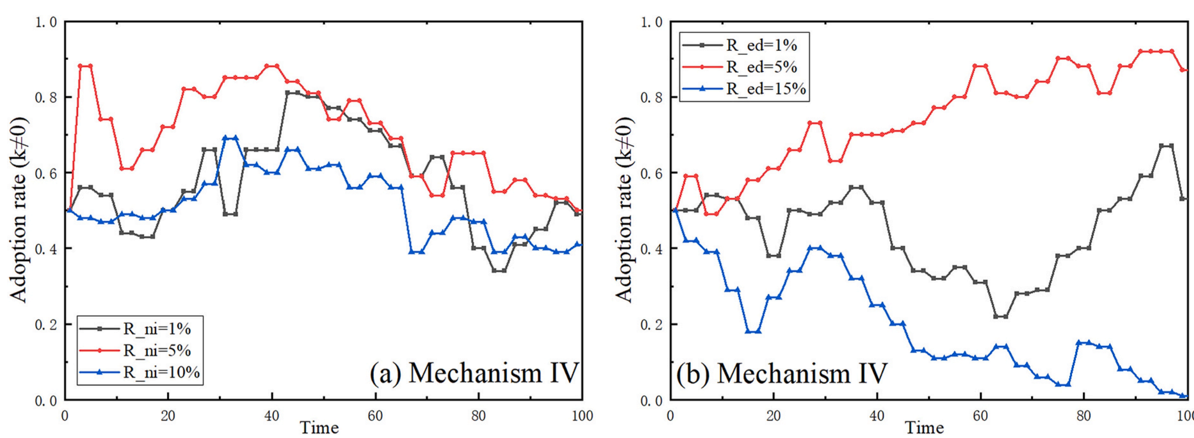

Fig. 14. Results of product diffusion under different R_{ni} and R_{ed}.

attribution bias [8], anchoring bias [9], and emotional bias [10]. There is scant research on collective overreaction in the field of consumer behavior. As prior research has shown, macro-evolution mechanisms of group-level irrational behavior emerge from individual interactions [32,55,56], which cannot be captured by prior research on individual-level irrational behavior. Therefore, our study extends these individual-level mechanisms to group-level mechanisms of irrational behavior. Specifically, by focusing on how individual interactions (i.e., preferential selection and imitation) drive collective overreaction, our study responds to the calls for research on understanding the "social influence" needed to successfully stimulate collective overreaction, as exemplified by Ye and Fan [57]. Moreover, our theorization sheds new light on the visualization of the diffusion process under different dynamic network growth mechanisms, especially intuitively showing the phenomenon of collective overreaction, making it possible to quantify and analyze collective irrational behavior.

Second, dynamic online product diffusion networks provide us with new insights into areas of consumer group behavior (e.g., collective overreaction). Prior studies on consumer collective irrational behavior have focused primarily on the static network perspective but have failed to gain insight into the impact of the dynamics of social networks. Our study extends these theoretical lenses to examine collective irrational behavior in product diffusion in a dynamic network context. Diverging from the research focusing primarily on the static analysis of consumer behavior under rational man assumptions, our study aims to open the black box and discover and comprehend collective irrational behavior in product diffusion from a dynamic, group-level perspective, thereby helping extend the consumer irrational behavior research paradigm. Our study breaks new theoretical ground for future research in this burgeoning area and potentially opens up new opportunities for the further investigation of consumer irrational behavior in information systems.

Third, this research framework helps visualize the evolution of collective overreaction under different dynamic network growth mechanisms, especially intuitively showing the phenomenon of collective overreaction. Drawing on a variety of well-established theories, we grasp the dynamic evolution characteristics of product diffusion over time in evolutionary games and explore the microscopic interactions among users under overreaction scenarios involving different network dynamic growth mechanisms. This approach makes it possible to quantify and analyze collective irrational behavior.

6.3. Practical application implications

Our findings have two significant practical implications for product diffusion. First, in network external dynamic growth mechanisms, increased KOL promotion combined with KOC promotion should be used in product marketing activities. To improve the overall product diffusion level, the focus should be on online communities with large nodes and large clustering coefficients. Moreover, the focus should be on KOL promotion in online communities with large nodes and KOC promotion in online communities with large clustering coefficients. KOL promotion helps expand the reach of advertising campaigns, while KOC promotion improves the reach efficiency of advertising campaigns. The key to the proliferation of products is to improve the connectivity and clustering of the network: with the high connectivity of the proliferation network, the number of product adopters will grow steadily; however, when the level of connectivity is not high, marketing campaigns can sometimes be counterproductive. Therefore, it is extremely important to use KOL promotion together with KOC promotion to expand and improve advertising reach and reach efficiency.

Second, in network internal dynamic growth mechanisms, when product promotion is in a bottleneck period, random recommendations can be adopted to increase the connectivity of individuals in the existing product diffusion network. Focus should be placed on the key nodes that receive low revenue from the product or provide negative comments about the product. In a network with a constant node size, improving the reach efficiency of advertising does not affect the level of product diffusion but has a positive effect on improving community stability.

6.4. Limitations and future research

This paper has certain limitations. First, our study did not consider directed networks. Second, our study did not account for individual differences among consumers. We will incorporate these elements in future research to simulate a more realistic phenomenon of collective overreaction.

CRediT authorship contribution statement

Xiaochao Wei: Writing – original draft, Project administration, Funding acquisition. **Yanfei Zhang:** Writing – original draft, Software, Methodology. **Xin (Robert) Luo:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

All authors disclosed no relevant relationships.

Data availability

Data will be made available on request.

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Appendix A. Appendix

Mechanism	Manipulated variable	Mean	Variance	p value of ANOVA	Mechanism	Manipulated variable	Mean	Variance	p value of ANOVA
I		−0.179	0.087	0.000	III	$P_{KOC} = 0.2$	0.003	0.042	0.006
II		−0.072	0.064		III	$P_{KOC} = 0.5$	0.003	0.037	
III		0.065	0.079		III	$P_{KOC} = 0.8$	0.026	0.009	
II	$z = 1\%$	0.072	0.081	0.002	III	$z = 1\%$	0.076	0.107	0.000
II	$z = 5\%$	0.040	0.080		III	$z = 5\%$	0.015	0.086	
II	$m_1 = 2$	−0.008	0.029	0.001	III	$m_1 = 2$	0.091	0.046	0.007
II	$m_1 = 5$	−0.031	0.079		III	$m_1 = 5$	0.060	0.061	
II	$m_1 = 8$	−0.035	0.065		III	$m_1 = 8$	0.026	0.047	
II	$R_{ni} = 5\%$	0.575	0.061	0.000	III	$R_{ni} = 5\%$	0.749	0.198	0.001
II	$R_{ni} = 10\%$	0.407	0.048		III	$R_{ni} = 10\%$	0.691	0.131	
II	$R_{ed} = 5\%$	0.559	0.417	0.000	III	$R_{ed} = 5\%$	0.869	0.220	0.007
II	$R_{ed} = 10\%$	0.400	0.416		III	$R_{ed} = 10\%$	0.672	0.126	
IV		−0.068	0.030	0.001	IV	$m_2 = 1\%$	−0.076	0.028	0.009
V		−0.073	0.129		IV	$m_2 = 3\%$	−0.093	0.029	
VI		−0.072	0.102		IV	$m_2 = 5\%$	−0.039	0.029	
V	$m_2 = 1\%$	−0.018	0.026	0.002	VI	$m_2 = 1\%$	−0.033	0.156	0.003
V	$m_2 = 3\%$	−0.002	0.111		VI	$m_2 = 3\%$	−0.047	0.028	
V	$m_2 = 5\%$	−0.073	0.108		VI	$m_2 = 5\%$	−0.007	0.081	
IV	$R_{ni} = 1\%$	0.495	0.128	0.000	IV	$R_{ed} = 1\%$	0.532	0.110	0.000
IV	$R_{ni} = 5\%$	0.521	0.118		IV	$R_{ed} = 5\%$	0.875	0.126	
IV	$R_{ni} = 10\%$	0.416	0.087		IV	$R_{ed} = 10\%$	0.014	0.130	

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Xiaochao Wei received the Ph.D. degree in Management Science and Engineering from the Huazhong University of Science and Technology, in 2013. He is currently a Professor of Economics at the Wuhan University of Technology. He has published research papers in leading journals including *Journal of the Association for Information Systems, Information Sciences, Journal of Artificial Societies and Social Simulation, Expert Systems with Applications, Scientific Programming*. His research interest includes management information systems simulation.

Yanfei Zhang received the M.E. degree in Applied Economics from the Wuhan University of Technology, in 2021, where she is currently pursuing the Ph.D. degree in Applied Economics. She has published research papers in leading journals including *Journal of the Association for Information Systems*. Her research interests include consumer behavior and multi-agent simulation.

Xin (Robert) Luo is a Special Assistant to the Dean for Research Advancement, an Endowed Dean's Professor of Research Excellence, and a Full Professor of Management Information Systems and Information Assurance at the Anderson School of Management of the University of New Mexico, Albuquerque, USA. He received his Ph.D. in Management Information Systems from Mississippi State University, USA. His research has been or will be published in leading IS/ business journals, including the *Information Systems Research*, *Journal of Operations Management, Production and Operations Management*, *Journal of Management Information Systems*, *Journal of the Association for Information Systems*, *European Journal of Information Systems, Information Systems Journal, Journal of Strategic Information Systems, Journal of Information Technology, Decision Sciences, Decision Support Systems, Information & Management*, and *IEEE Transactions on Engineering Management*. He has served as an ad hoc Associate Editor for *MIS Quarterly* and an Associate Editor for the *European Journal of Information Systems*, and currently serves as an Associate Editor for the Decision Support System, *Journal of the Association for Information Systems, Decision Sciences, Information & Management, Electronic Commerce Research*, and the *Journal of Electronic Commerce Research*. He also sits on the Editorial Review Board of Information Systems Research and serves as a Guest AE for special issues of *Journal of Management Information Systems and Decision Support Systems*. His research interests center around information assurance, innovative technologies for strategic decision-making, and global IT management. He is the Co-Editor-in-Chief of the *International Journal of Accounting and Information Management*.